

 Multimodal Time-series Corpus

Task 1 Electricity Consumption

This is the electricity consumption recorded in Kilowatt (kW) in **city A**. Scenario: Suppose that there is a heat wave in **city A** from 2012-10-09 18:00:00 for 3 hours, leading to excessive use of air conditioning, and 5 times the usual electricity being consumed.

Task 2 Temperature Sensor

A temperature **sensor** in **city A** malfunctioned from 2025-10-01 00:00 to 2025-10-06 11:00, causing abnormal readings.

Task 3 Water Demand

Daily water demand in **city A** was much higher during the heat wave from 2012-10-09 ...

1 Entity Extraction LLM

- Extract type-level entities
- One primary entity per task
- Align primary entity with the time series variable
- Extract explicit contain relations

Entities			Relations	
Entity	Tasks	Primary Entity	Parent	Child
City	1, 2, 3	Task 1, Task 3	City	Sensor
Sensor	2	Task 2	...	...

Entity Judge

One primary entity
Align with TS variable
Correct relations

Human in the loop



2 Profile Generation LLM

- Compute minimal profile count K
- Generate K synthetic, realistic profiles

City Profile
Name: Veridian
Type: Urban Area, Mid-sized
Population: ~250,000
Description: ...



Sensor Profile
Name: ThermoLink XT-400
Type: Precision Thermometer
Measurement: Temperature
Unit: degree Celsius (°C)
Accuracy: ± 0.5 °C



Minimal Profile Generation
Maximum Profile Reuse

3 Profile Assignment LLM

- Assign one profile for the primary entity
- Include ancestor profile if applicable
- Rewrite entity profile only

Original Context

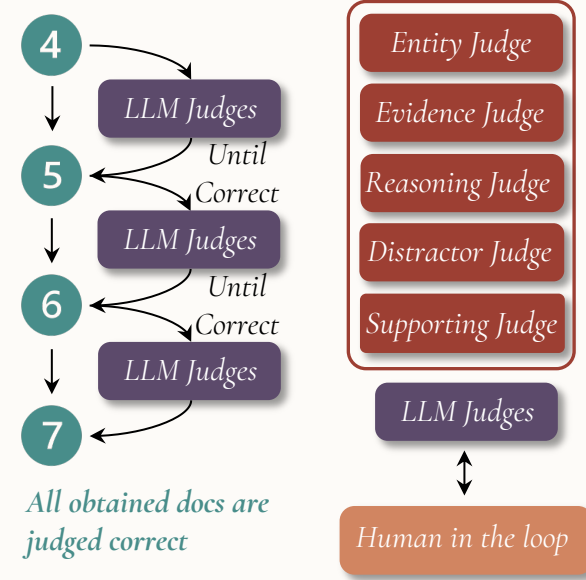
This is the electricity consumption recorded in Kilowatt (kW) in **city A**. Suppose that there is a heat wave in **city A** from 2012-10-09 18:00:00 for 3 hours, leading to excessive use of air conditioning, and 5 times the usual electricity being consumed.

Rewritten Context

This is the electricity consumption recorded in kW in **Veridian city**. Suppose that there is a heat wave in **Veridian city** from 2012-10-09 18:00:00 for 3 hours, leading to excessive use of air conditioning, and 5 times the usual electricity being consumed.

Agent Judge

TS consistency: TS & context consistency
Wrong number: Conflicts with real data.
Too precise: Adds fake precision.
Out of scale: Unrealistic numbers.
Hallucination: Wrong contexts



4 Evidence Extraction LLM

- Preserve all details and facts
- Collectively exhausted
- Extract causal relationships
- Extract associated time periods

Rewritten Context

This is the electricity consumption recorded in Kilowatt (kW) in Veridian city. There is a **heat wave** in Veridian city from **2012-10-09 18:00:00 for 3 hours**, leading to **excessive use of air conditioning**, and **5 times the usual electricity** being consumed.

Supporting Evidences

E1: Suppose that there is a **heat wave** in Veridian Heights from **2012-10-09 18:00:00 for 3 hours**
E2: leading to **excessive use of air conditioning**, and **5 times the usual electricity** being consumed

Evidence Judge

Coverage
Time consistency
Details

Human in the loop



5 Reasoning Chain Expansion LLM

- Add reasoning hops (controllable)
- No branch, no cycle, no shortcuts
- Evidence rewrite

Reasoning Trace

Reasoning Evidence 1: On the **ninth of October in 2012**, the climatic conditions within the Veridian Heights area transitioned into an episode of **extreme thermal elevation**.

Reasoning Link 1: The heat wave compelled residents to **activate power-intensive climate management machinery simultaneously**.

Reasoning Evidence 2: The deteriorating conditions inside the structures created a simultaneous and widespread necessity for people to utilize high-power environmental regulation systems.

Reasoning Link 2: This reliance on cooling systems throughout the evening caused the power grid's throughput to increase **fivefold**.

Reasoning Evidence 3: The massive use of cooling appliances for the **180-minute window starting at 6:00 PM** resulted in the regional electrical consumption reaching **500% of its standard load**.

Controllable Difficulty (Reasoning hops)



Reasoning Judge

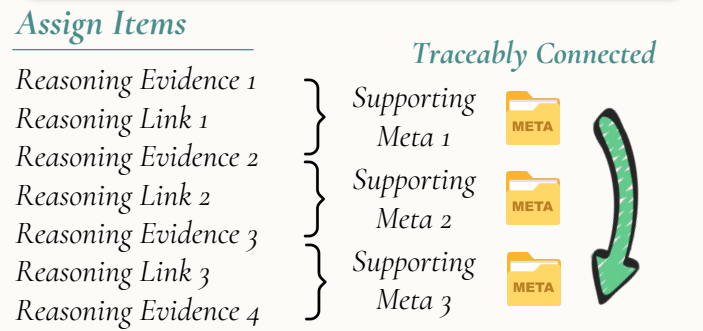
No reasoning shortcut
Details coverage
Natural reasoning trace

6 Supporting Meta Generation LLM

Assign items
2RE + connecting RL

Timeseries

Supporting Metadata Generation



Supporting Meta Generation

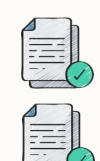
Name: Heat Wave Electricity Consumption Notice
Type: Utility Event Notice
Issuer: City Energy Monitoring Office
Provides: heat wave on 2012-10-09 18:00:00 for 3 hours; excessive AC associated with 5x usual electricity.
Length: 140 words

Supporting Doc Judge

Coverage
Inference realism
Context alignment

Temporal consistency
Entity consistency
Collective usefulness

Supporting Documents



7 Distractor Meta Generation LLM

Distractor Drafting

Timeseries

Distractor Metadata Generation

Distractor Taxonomy

Confounder: Plausible but false cause; points to the wrong target-variable outcome.
Noisy: Related-looking but irrelevant details; adds no predictive signal.
Profile: Same event type, wrong profile → the wrong future for the target variable.
Temporal: Wrong timing; event occurs contradicts the true future.
Timeseries: Misreads the observed history and projects that error into the future.

Timeseries Distractor Taxonomy

point × value: Misread one data point or local extremum → wrong future value.
segment × value: Misread a local segment's level → wrong future level.
segment × trend: Misread a local segment's direction → wrong future direction.
segment × periodicity: Misread local cycles or frequency → wrong future cycles.
global × value: Misread the overall level → wrong overall future level.
global × trend: Misread the overall trend → wrong long-run direction.
global × periodicity: Misread the overall cycle pattern → wrong long-run cyclicity.

Supporting Meta

Distractor Meta

8 MetazDoc Generation

Distractor Documents



Distractor Doc Judge

Realism
Distinguishability
Relevance
Type consistency